

Evaluating AWS DocumentDB Serverless for Cost Optimization

Amazon DocumentDB serverless is an on-demand, auto scaling configuration that automates the processes of dynamically adjusting the capacity of Amazon DocumentDB database based on application demand. Charging only for the resources that clusters consume.

Deployment Options

Amazon DocumentDB offers different deployment options to cater to various workload patterns and operational preferences.

	Provisioned Clusters	Elastic Clusters	Serverless <i>Launched and made generally available on July 31, 2025</i>
Scalability	Requires manual scaling. Scaling up or down involves modifying instance classes, which can lead to brief outages.	Designed for massive scale-out, handling millions of reads and writes per second. It distributes data across multiple shards, allowing for horizontal scaling of throughput.	Automatically scales compute capacity up or down in fine-grained increments based on real-time application demand. This eliminates the need for manual capacity planning.

Cost Model	Predictable, based on the chosen instance class and storage provisioned. Suitable for steady, predictable workloads where resource requirements are relatively constant.	Combines aspects of both provisioned and serverless, offering a balance between predictable capacity and dynamic scaling for high-throughput workloads.	Pay-per-use, billing for DocumentDB Capacity Units (DCUs) consumed, which are a combination of CPU and memory. Ideal for unpredictable or spiky workloads where provisioning for peak capacity would lead to idle resources and higher costs.
Operational Overhead	Higher, as it requires careful capacity planning and manual adjustments to meet changing workload demands.	Requires understanding of sharding and distributed database concepts, but simplifies the management of large-scale, sharded deployments compared to self-managed solutions.	Low, as AWS manages the scaling and underlying infrastructure.
Use Cases	Production applications with stable and predictable performance requirements, where consistent performance and	Extremely high-throughput applications, large-scale IoT data processing, real-time analytics, and other scenarios demanding extreme	Websites and applications with variable workloads, development/testing environments, and scenarios where cost

	reliability are prioritized over dynamic cost optimization.	performance and scalability beyond what a single provisioned cluster can offer.	optimization through automatic scaling is paramount.
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Advantages of Amazon DocumentDB serverless

1. Simpler capacity management than provisioned.
2. Faster and easier scaling during periods of high activity.
3. Cost-effective during periods of low activity
4. Feature parity with provisioned instances.

Use Cases

The unit of measure for Amazon DocumentDB serverless is the DocumentDB Capacity Unit (DCU). Each DCU is a combination of approximately 2 gibibytes (GiB) of memory, corresponding CPU, and networking. DocumentDB serverless is supported on DocumentDB 5.0.0 and higher with a capacity range of 0.5 - 256 DCUs.

- Production environments hosting business-critical applications that require high scale and availability. For a provisioned cluster, scaling up requires adding a whole new instance. DocumentDB serverless can add half a DCU when only a little more capacity is needed. It can add 0.5, 1, 1.5, 2, or additional half-DCUs based on the additional capacity needed to handle an increase in workload. And it can remove 0.5, 1, 1.5, 2, or additional half-DCUs when the workload decreases and that capacity is no longer needed.
- Development and testing environments with a low minimum capacity instead of using burstable db.t* instance classes. When the database isn't in use, all of the instances scale down to avoid unnecessary charges. In order to support [scaling down to 0.5 DCUs when idle](#), instance limits are capped if the MinCapacity is configured to be less than or equal to 1.0 DCUs.

Database capacity can be adjusted at any time by modifying the existing instances from provisioned to DocumentDB serverless. No need to create a new cluster or a new instance.

Monitoring Provisioned instances

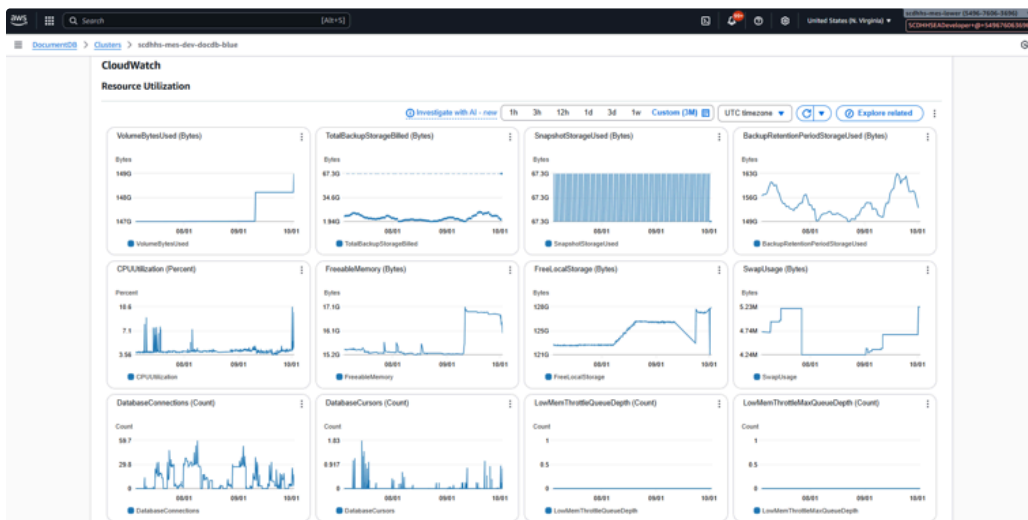
Currently in mes-lower account there are 3 DocumentDB clusters with a single provisioned instance with a size of db.r6g.2xlarge for Development and db.r6g.4xlarge for QA environment.

Cluster identifier	Status	Role	Engine version	Region & AZ	Instance health	CPU	Current activity	Size	Storage type	Maintenance
scdhhs-mes-dev-docdb-blue	Available	Regional cluster	5.0.0	us-east-1	Healthy	5.26%	13 Sessions	1 Instance	I/O-Optimized	Available
docdbinstance1-e2z9h9y9e1	Available	Primary instance	5.0.0	us-east-1c	Healthy	5.26%	13 Sessions	db.r6g.2xlarge	-	Available
scdhhs-mes-dev-docdb-green	Available	Regional cluster	5.0.0	us-east-1	Healthy	4.21%	6 Connections	1 Instance	I/O-Optimized	Available
docdbinstance1-e2z9h9y9e1	Available	Primary instance	5.0.0	us-east-1b	Healthy	4.21%	6 Connections	db.r6g.2xlarge	-	Available
scdhhs-mes-qa-docdb	Available	Regional cluster	5.0.0	us-east-1	Healthy	2.08%	0 Connections	1 Instance	I/O-Optimized	Available
docdbinstance1-e2z9h9y9e1	Available	Primary instance	5.0.0	us-east-1b	Healthy	2.08%	0 Connections	db.r6g.4xlarge	-	Available

The CPU, RAM, disk space, network bandwidth resources utilized in the last 3 months

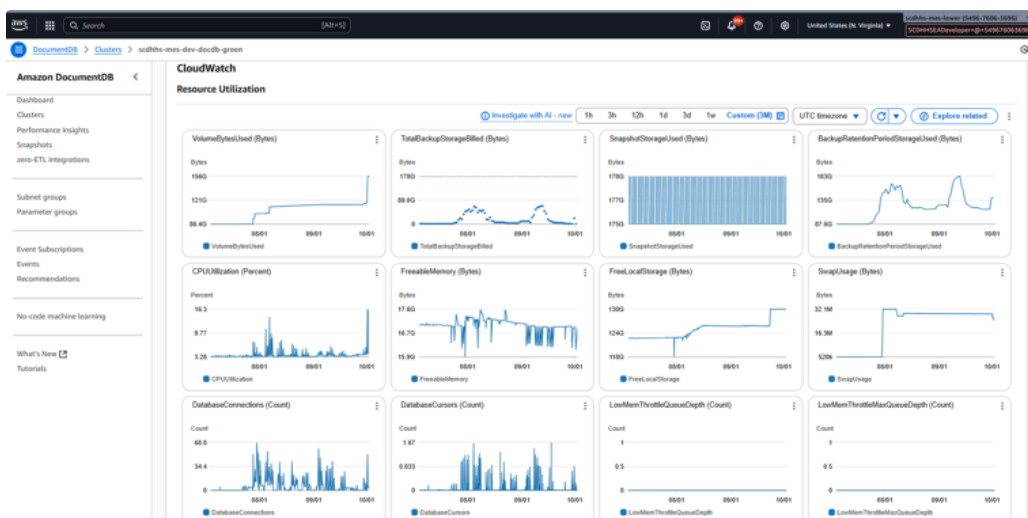
1. Dev Blue → db.r6g.2xlarge (8 vCPUs, 64 GB of RAM)

<https://us-east-1.console.aws.amazon.com/docdb/home?region=us-east-1#cluster-details/scdhhs-mes-dev-docdb-blue>



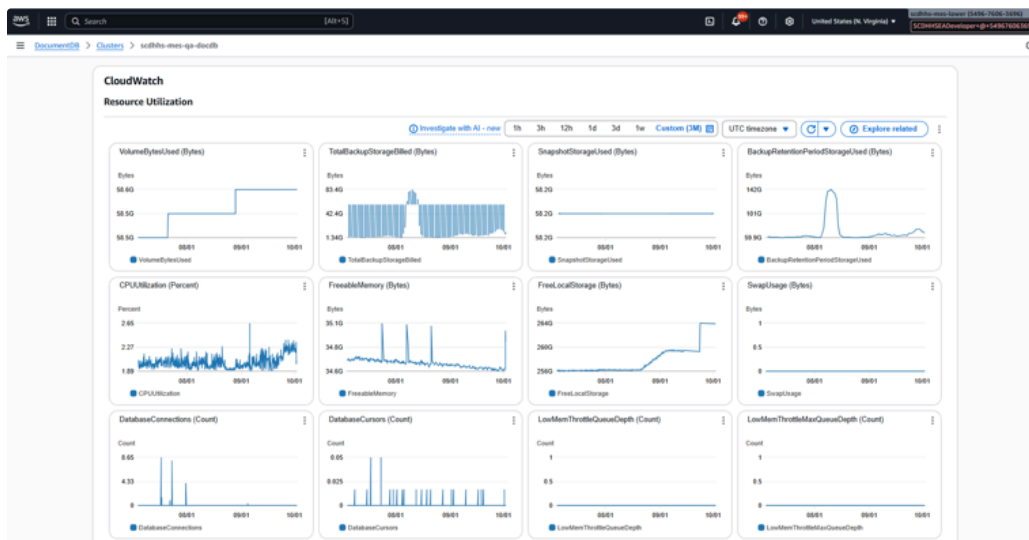
2. Dev Green → db.r6g.2xlarge (8 vCPUs, 64 GB of RAM)

<https://us-east-1.console.aws.amazon.com/docdb/home?region=us-east-1#cluster-details/scdhhs-mes-dev-docdb-green>



3. QA → db.r6g.4xlarge (16 vCPUs and 128GB of RAM)

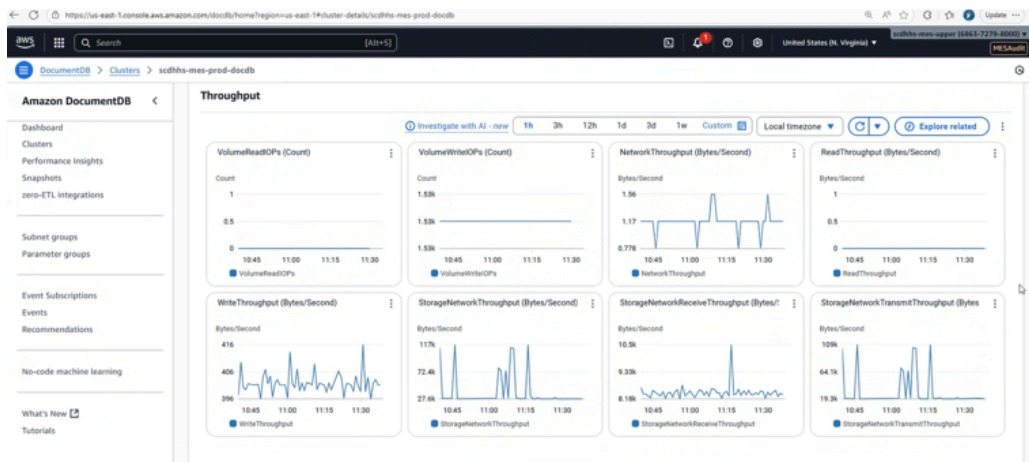
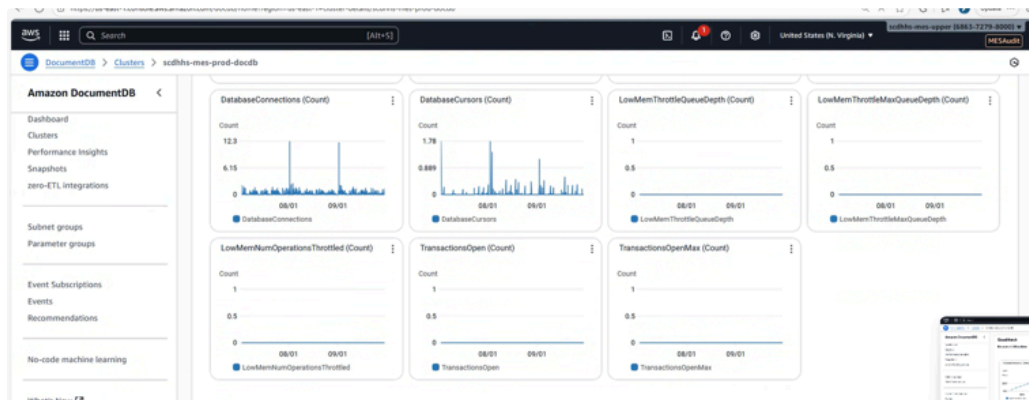
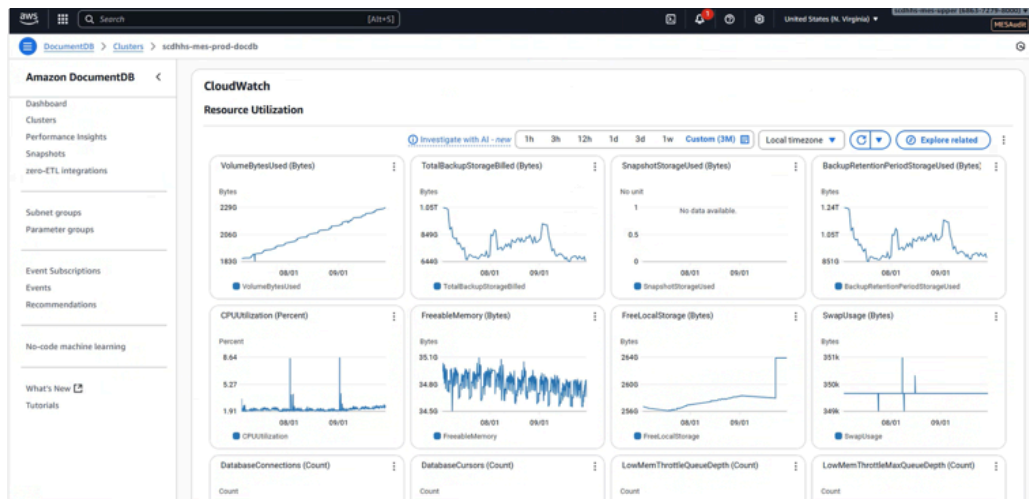
<https://us-east-1.console.aws.amazon.com/docdb/home?region=us-east-1#cluster-details/scdhhs-mes-qa-docdb>



In mes-upper account, there are two DocumentDB clusters with a multi-node provisioned instances with a size of db.r6g.4xlarge for Staging and Production environments. Each cluster contains a single Writer and two Reader (Replica) nodes for high availability. Serverless reader instances can be also used to offload some of the read-only workload from the writer instance however the existing Glue jobs and Lambda functions workloads are only using the Cluster DNS endpoint for writing and reading operations.

Cluster identifier	Status	Role	Engine version	Region & AZ	Instance health	CPU	Current activity
scdhhs-mes-prod-docdb	Available	Regional cluster	5.0.0	us-east-1	-	-	-
docdbinstance1-1eqhbgfrcf	Available	Replica instance	5.0.0	us-east-1a	Healthy	2.17%	0 Connections
docdbinstance1-1qghffgflm	Available	Primary instance	5.0.0	us-east-1b	Healthy	2.74%	0 Connections
docdbinstance2-stmrnb2o6f9d	Available	Replica instance	5.0.0	us-east-1c	Healthy	2.19%	0 Connections
scdhhs-mes-qa-docdb	Available	Regional cluster	5.0.0	us-east-1	-	-	-
docdbinstance1-vevjdyk2kxur	Available	Replica instance	5.0.0	us-east-1c	Healthy	2.33%	0 Connections
docdbinstance1-duh3ymovv52	Available	Primary instance	5.0.0	us-east-1b	Healthy	2.88%	0 Connections
docdbinstance1-gr0hmuhf9pgo	Available	Replica instance	5.0.0	us-east-1a	Healthy	2.44%	0 Connections

Instance health	CPU	Current activity	Size	Storage type	Maintenance	VPC	Creation Time
-	-	-	3 Instances	I/O-Optimized	Available	-	8/8/2024, 7:18:03 PM UTC-4
Healthy	2.11%	0 Connections	db.r6g.4xlarge	-	Available	vpc-0e0b0e4...	8/8/2024, 7:29:44 PM UTC-4
Healthy	2.66%	0 Connections	db.r6g.4xlarge	-	Available	vpc-0e0b0e4...	8/8/2024, 7:23:48 PM UTC-4
Healthy	2.16%	0 Connections	db.r6g.4xlarge	-	Available	vpc-0e0b0e4...	8/8/2024, 7:30:32 PM UTC-4
-	-	-	3 Instances	I/O-Optimized	None	-	7/30/2024, 8:16:24 AM UTC-4
Healthy	2.22%	1 Sessions	db.r6g.4xlarge	-	Available	vpc-029ab19c...	7/30/2024, 8:21:20 AM UTC-4
Healthy	2.75%	1 Sessions	db.r6g.4xlarge	-	Available	vpc-029ab19c...	8/9/2024, 5:08:44 PM UTC-4
Healthy	2.26%	1 Sessions	db.r6g.4xlarge	-	Available	vpc-029ab19c...	8/14/2024, 3:05:09 PM UTC-4



Given the analysis of current performance metrics, there are [cost optimization opportunities](#) for tuning the provisioned instances by using serverless capabilities.

Monitoring serverless scaling

The ServerlessDatabaseCapacity and DCUUtilization CloudWatch metrics helps to determine how much capacity the database is actually using and where that capacity falls within the specified range. For more information, see [Monitoring Amazon DocumentDB serverless](#).

DocumentDB serverless scaling can happen while database connections are open, while transactions are in process, etc. DocumentDB serverless doesn't wait for a quiet point to begin scaling. Scaling doesn't disrupt any database operations that are underway.

Serverless scaling configuration

The `ServerlessV2ScalingConfiguration` argument specifies the scaling capacity range of your DocumentDB serverless instances. It consists of the [minimum and maximum DocumentDB capacity unit \(DCU\) values](#) that apply to all the DocumentDB serverless instances in the cluster

Choosing the MinCapacity setting

AWS recommends setting the minimum to a value that allows each writer or reader to hold the working set of the application in the buffer pool. That way, the contents of the buffer pool aren't discarded during idle periods.

If `MinCapacity` is set to less than or equal to 1, then the limit values for cursor limit, open transactions, and connections (active) are capped to a lower maximum value. The technical reason for the cap is to support scaling down to an “idle” state. If these capped limits are insufficient for the workload, use a `MinCapacity` value of at least 1.5 DCUs.

In the idle state, the DocumentDB serverless instances do not have sufficient CPU compute capacity to support most production workloads but are ready to quickly scale up to support a new workload. In a non-idle state, DocumentDB serverless instances typically require at least 1.0 - 2.5 DCUs. Therefore, when DocumentDB serverless instances scale up from an idle state to a non-idle state, they will scale up directly to 1.0 - 2.5 DCUs (or the value of `MaxCapacity` if it is lower).

If we have a cluster with DocumentDB serverless reader instances, AWS recommends setting the minimum capacity to a value that represents a comparable amount of memory and CPU to the writer instance.

db.serverless instance limits if MinCapacity ≤1

Current Capacity (DCU)	Instance Memory (GiB)	Connections (all)	Cursor Limit	Open Transactions	Connections (active)
0.5	1	250	6	12	39
1	2	500	12	24	79
2	4	1000	24	48	173
4	8	2000	48	96	416
8	16	4000	96	192	1071
16	32	8000	132	264	1550
32	64	16000	132	264	1550
64	128	32000	132	264	1550
128	256	60000	132	264	1550
256	512	60000	132	264	1550

db.serverless instance limits if MinCapacity >1

Current Capacity (DCU)	Instance Memory (GiB)	Connections (all)	Cursor Limit	Open Transactions	Connections (active)
1.5	3	7250	18	36	124
2	4	1000	24	48	173
4	8	2000	48	96	416
8	16	4000	96	192	1071
16	32	8000	192	384	2709
32	64	16000	384	768	4500
64	128	32000	768	1536	4500
128	256	60000	1536	3072	4500
256	512	60000	3072	6144	4500

In particular, AWS recommends the following minimum **MinCapacity** for use with the specified features (these recommendations are subject to change):

- Performance Insights — 2 DCUs

The time it takes for a DocumentDB serverless instance to scale from its minimum capacity to its maximum capacity depends on the difference between its minimum and maximum DCU values. When the current capacity of the instance is large, DocumentDB serverless scales up in larger increments than when the instance starts from a small capacity.

Choosing the MaxCapacity setting

In particular, AWS recommend the following minimum **MaxCapacity** for use with the specified features (these recommendations are subject to change):

- Serverless instance creation on a cluster with a large data volume – 2 DCUs (this includes serverless instance creation as part of a cluster restore.)
- Certain instance limits are determined by the instance's current capacity, such as connections limit, cursor limit, and open transactions limit. When choosing the **MaxCapacity** value for the workload, make sure to keep these instance limits in mind in order to avoid being bottlenecked by one of these limits.

It's recommended the monitoring of the scaling and resource usage of serverless instances to detect if a serverless instance is frequently scaling to maximum capacity and hitting resource constraints. It's also important to avoid out-of-memory errors since DocumentDB serverless instances could reach the limit of its maximum capacity and will set the instance to a status of incompatible-parameters. While the instance has the incompatible-parameters status, some operations are blocked.

Promotion tier of serverless readers

For serverless instances, the tier number also determines whether the instance scales up to match the capacity of the writer instance or scales independently based on its own workload. Serverless reader instances in tier 0 or 1 are kept at a minimum capacity at least as high as the writer instance. That way, they are ready to take over from the writer instance in case of a failover.

DocumentDB serverless reader instances in tiers 2–15 don't have the same constraint on their minimum capacity. When they are idle, they can scale down to the minimum DocumentDB capacity unit (DCU) value specified in the cluster's capacity range.

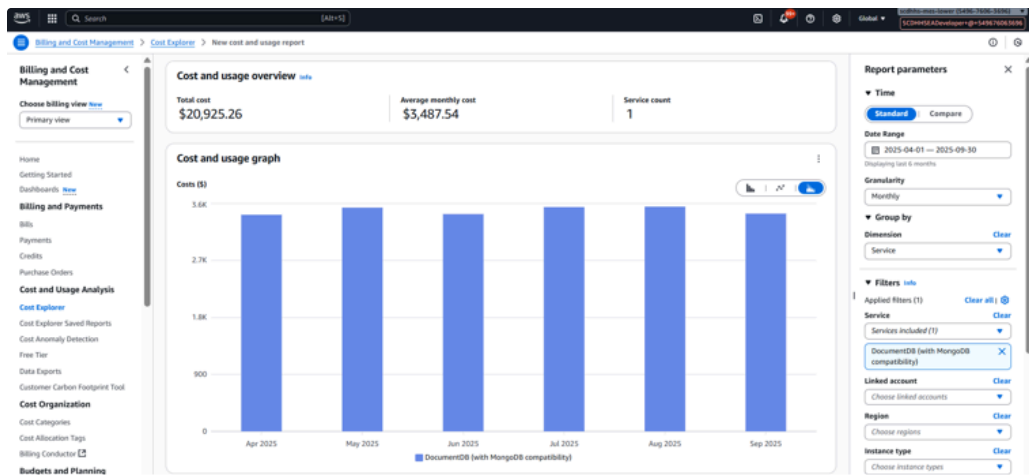
If we have a cluster with DocumentDB serverless reader instances, the readers don't scale along with the writer instance when the promotion tier of the readers isn't 0 or 1. In that case, consider that setting a low minimum capacity can result in excessive replication lag.

Serverless pricing

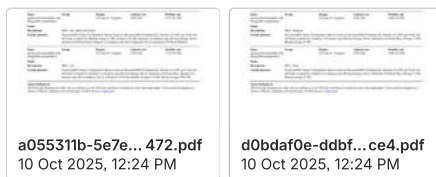
The charges for DocumentDB serverless capacity are measured in terms of DCU-hours. For information about how DocumentDB serverless charges are calculated, see [Amazon DocumentDB pricing](#)

Region:		
US East (N. Virginia) ▼		
Measure	Amazon DocumentDB Standard (per DCU hour)	Amazon DocumentDB I/O-Optimized (per DCU hour)
DocumentDB Capacity Unit	\$0.0822	\$0.0905
Starting capacity	0.5 DCU	
Capacity granularity	Fine-grained increments of 0.5 DCU	
Read replicas	Up to 15 Amazon DocumentDB Replicas replicas for read scalability	
Multi-AZ and SLA	Multi-AZ and SLA: Distribute read replicas in separate AZs for high availability. You can refer to Amazon DocumentDB SLA for details.	

It's possible to calculate and compare with existing billing of monthly charges in lower and upper accounts.



Also, using the AWS Calculator the estimated monthly charges are roughly \$14,400 considering MES lower and upper environments.



Migrating to Amazon DocumentDB serverless

In our case to switch a provisioned cluster to use DocumentDB serverless, follow these [steps](#). It's also possible to add one or more DocumentDB serverless instances to the existing cluster as reader instances. Using the DocumentDB failover mechanism to promote an DocumentDB serverless instance to be the writer and check how it handles the read/write workload. That way, it's feasible to switch over with minimal downtime and without changing the endpoint that the client applications are using.

Suggested and Calculated Scaling Configuration

It's important to [monitor the AWS DocumentDB serverless instances](#) to avoid Out of Memory: incompatible-parameters status. Also, the following AWS CloudWatch metrics for DocumentDB serverless needs active monitoring to understand how the DocumentDB serverless instances are scaling up and down:

- ServerlessDatabaseCapacity
- DCUUtilization
- CPUUtilization
- FreeableMemory
- TempStorageIops

- TempStorageThroughput

For upper environments, it's recommended to enable [Performance Insights](#) to monitor the performance of DocumentDB serverless instances.

AWS Environment	Current Provisioned Instances	Min Capacity	Max Capacity
Development	db.r6g.2xlarge (8 vCPUs, 64 GB RAM)	0.5 - 2 DCU's	16 - 32 DCU's
QA	db.r6g.4xlarge (16 vCPUs, 128 GB RAM)	0.5 - 2 DCU's	16 - 32 DCU's
Preprod	db.r6g.4xlarge (16 vCPUs, 128 GB RAM)	2 - 4 DCU's	32 - 64 DCU's
Production - Reader Production - Primary	db.r6g.4xlarge (16 vCPUs, 128 GB RAM)	4 - 8 DCU's 4 - 8 DCU's	32 - 64 DCU's 32 - 64 DCU's

Estimated Cost Savings

Considering the DCU pricing per hour of \$0.0905 with an average usage of 25% per month, the estimated charges are displayed in the following table

Environment	Average Usage 25% per month	Monthly charges
Dev	$32 * 0.25 = 8 \text{ DCU}$	$8 * 0.0905 * 24 * 30 = \$521.28$
QA	$32 * 0.25 = 8 \text{ DCU}$	$8 * 0.0905 * 24 * 30 = \$521.28$
Preprod	$64 * 0.25 = 16 \text{ DCU}$	$16 * 0.0905 * 24 * 30 = \$1,042.56$

Prod - Reader	64* 0.25 = 16 DCU	16 * 0.0905 * 24 * 30 = \$1,042.56
Prod - Primary	64* 0.25 = 16 DCU	16 * 0.0905 * 24 * 30 = \$1,042.56

The estimated total monthly charges with Serverless instances would be \$4,170.24 rather than \$14,400, providing a potential 70% of monthly cost savings.

Steps to Change DocumentDB to Serverless in CloudFormation

1. Modify the DocumentDB Cluster Resource:

Update the [AWS::DocDB::DBCluster](#) resource in the CloudFormation template to include the `ServerlessV2ScalingConfiguration` property. This property defines the minimum and maximum capacity units (DCUs) for the serverless cluster.

```

1 Resources:
2     MyDocDBCluster:
3         Type: AWS::DocDB::DBCluster
4         Properties:
5             # ... other cluster properties ...
6             ServerlessV2ScalingConfiguration:
7                 MinCapacity: 0.5 # Example: minimum 0.5 DCUs
8                 MaxCapacity: 16 # Example: maximum 16 DCUs

```

2. Modify DocumentDB Instance Resources:

If you have existing [AWS::DocDB::DBInstance](#) resources, you will need to modify them to specify the `db.serverless` instance class.

```

1 Resources:
2     MyDocDBInstance:
3         Type: AWS::DocDB::DBInstance
4         Properties:
5             # ... other instance properties ...
6             DBInstanceClass: db.serverless
7             DBClusterIdentifier: !Ref MyDocDBCluster

```

Also, add some additional CloudWatch alarms for the following metrics

- ServerlessDatabaseCapacity
- DCUUtilization
- TempStorageIops
- TempStorageThroughput